

Validity Analysis: MRBENCH

Target context: North India Grades 6–8 Mathematics Students (AI Tutoring)

Overall risk: **HIGH**

Deployment Context

Use case: Deployment scenario: With this benchmark, the user is trying to assess the pedagogical soundness of the LLMs acting as AI tutors for student mistake remediation tasks in mathematics. Domain: AI for Education

Target population: Students in Grades 6–8 in North India receiving one-on-one mathematics mistake-remediation tutoring. Instruction is conducted in Indian English, with many learners being bilingual in Hindi and English.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	medium
Input Content ■	1	Serious concern	high
Input Form ■	2	Significant gaps	high
Output Ontology	2	Significant gaps	medium
Output Content ■	2	Significant gaps	high
Output Form ✓	3	Moderate gaps	medium
Average	2.2		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MRBench provides a well-motivated, task-focused evaluation framework for LLM tutor responses in mathematics mistake remediation, with a defensible eight-dimension taxonomy [Q1, Q15], rigorous in-house annotation [Q32–Q38, Q60], and thoughtful metric design that rejects surface NLG proxies [Q4, Q7]. For the North India Grades 6–8 deployment, however, validity is substantially compromised on multiple dimensions. Input Content is the most severe failure: MRBench's Western-origin problems (Bridge, MathDial) contain no NCERT-aligned content, no Indian numeral conventions (lakh/crore) that are formally taught from Class 6 [WEB-7, WEB-8], no rupee word problems, and no Indian English / Hinglish / Devanagari / Romanized-Hindi input [Q51; WEB-1, WEB-19]. Input Form and Output Form share the English-only, text-only,

single-turn limitations, with independent evidence (Indi-RomCoM [WEB-1]) that LLMs degrade on Romanized code-mixed inputs — so English-only evaluation likely overstates deployment performance. Output Ontology and Output Content encode Western pedagogical norms and rest on a homogeneous CS-postgraduate annotator pool with no Indian teaching context [Q30, Q31]; culturally contested defaults (e.g., 'not revealing the answer' [Q22]) are not validated against Indian stakeholders. Input Ontology is the least compromised — the task family is correct — but the taxonomy omits code-switching, curriculum-fit, and cultural authority/tone categories. Regulatory context (DPDP Act 2023, sector-wide non-compliance [WEB-11, WEB-13]) further raises the stakes of unvalidated child-facing deployment.

ID	Evidence
Q1	"We propose a unified evaluation taxonomy with eight pedagogical dimensions based on key learning sciences principles, which is designed to assess the pedagogical value of LLM-powered AI tutor responses grounded in student mistakes or confusions in the mathematical domain." (p.1)
Q4	"General domain-agnostic natural language generation (NLG) metrics are not well-suited for this context, as most of them fail to account for pedagogical values and require gold references, which are often not available, especially in online interactions." (p.1)
Q7	"General domain-agnostic natural language generation (NLG) metrics like BLEU (Papineni et al., 2002), BERTScore (Lin, 2004), DialogRPT (Gao et al., 2020), and so on have been used as proxies to measure the coherence and human-likeness of AI tutor responses." (p.2)
Q15	"In this section, we first present our approach, narrowing the evaluation taxonomy down to eight measurable dimensions aligned with key pedagogical strategies (Jurenka et al., 2024; Hennessy et al., 2016). These dimensions are most suitable for the student mistake remediation task and are based on the learning sciences principles." (p.3)
Q22	"Since most dialogues are relatively short and present contexts for the mistakes made early in the student's solution, a good tutor strategy is not to reveal the answer to the student immediately but rather provide helpful guidance." (p.4)
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)
Q31	"We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient." (p.5)
Q32	"To control the annotation workflow and ensure quality, we opted not to use public annotation outsourcing platforms such as Prolific or MTurk, which allowed us to implement rigorous training protocols and a robust validation mechanism for the annotations." (p.5)
Q33	"First, we provided all annotators with comprehensive training, including an interactive training document (see Section C for more details) and oral instructions." (p.5)
Q34	"Following this, we conducted validation pilot study to evaluate the annotation quality and the annotators' understanding of the instructions before rolling out the large-scale human evaluation detailed in Section 5.2." (p.5)
Q35	"In this validation pilot study, all four annotators iteratively reviewed the annotation scheme and guidelines." (p.5)
Q36	"Each annotator also independently labeled the same eight randomly sampled dialogues – four from each of the two datasets (Bridge and MathDial) – across the eight dimensions of the evaluation taxonomy." (p.5)
Q37	"Given that each dialogue contained multiple responses from both LLMs and humans, and each response was annotated across eight evaluation dimensions, this resulted in a total of 544 annotations per annotator." (p.5)
Q38	"To measure inter-annotator agreement, we computed Fleiss' kappa value, which for this annotation experiment equals 0.65, indicating substantial agreement." (p.5)
Q51	"For the 192 instances in MRBench (60 from Bridge and 132 from MathDial), we generated appropriate subsequent responses based on the conversation history and the last utterance, which contained confusions or mistakes, using seven state-of-the-art LLMs." (p.6)
Q60	"The annotators reached an average Cohen's kappa score of 0.71, which indicates substantial inter-annotator agreement (McHugh, 2012)." (p.6)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench's output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-7	NCERT Class 6 numeral conventions differ from MRBench; annotators would need Indian curricular familiarity to judge notation-related responses (https://edusaksham.com/chapters/CBSE-Class-6-Mathematics-Numeration-Systems-Notes.html)
WEB-8	NCERT Class 7 Ganita Prakash confirms lakh/crore use in problems (https://edurev.in/t/407589/ncert-solutions-class-7-mathematics-ganita-prakash-large-numbers-around-us)
WEB-11	https://www.matters.ai/compliance/dpdp/dpdp-act-2023

WEB-13	71% of AI platforms used by minors in India are DPDP-non-compliant — regulatory context increases importance of Indian-stakeholder-validated labels for child-facing AI (https://asiaresearch.co/dpdp-compliance-in-respect-to-children-data/)
WEB-19	HiACC corpus documents Hindi–English code-switching in children's communication contexts (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12329218/)

Practical Guidance

What This Benchmark Measures

MRBench measures whether English-only LLM tutor responses to student mistakes in Western-origin elementary and middle-school mathematics dialogues receive desired categorical labels on eight Western-learning-sciences-grounded pedagogical dimensions [Q1, Q15, Q63], as judged by CS-postgraduate annotators [Q30]. Strongest dimensions for the deployment are Input Ontology (task-family match) and Output Form (defensible metric design over surface NLG [Q4]).

Construct Depth

Construct depth is moderate on the mistake-remediation construct itself: eight dimensions, three-tier labels, per-dataset splits [Q57, Q145, Q146], and substantial inter-annotator agreement [Q38, Q60] make it a genuinely diagnostic instrument for English-language pedagogical quality on Western-origin problems. Depth is shallow on culturally, linguistically, and curricularly localized constructs: MRBench cannot evaluate code-switching handling, curriculum-fit, Indian-numeral notation reasoning, cultural authority-tone calibration, multi-turn dialogue quality [Q117, Q118], or learning gains. Output-content depth is limited by the homogeneous annotator pool [Q30, Q31].

What Else You Need

For a defensible deployment decision, MRBench must be supplemented by: (1) a curriculum-aligned mistake-remediation set drawn from NCERT/CBSE Grades 6–8 problems including Indian numeral conventions and rupee word problems (addresses IC, IO); (2) a code-mixed / Hinglish input evaluation, potentially building on Indi-RomCoM's GSM8K-derived task [WEB-1] (addresses IF, OF); (3) re-annotation of a MRBench subset by Indian teachers and Grade 6–8 students to test convergent validity of desired labels, especially tone, revealing-the-answer, and motivation dimensions (addresses OO, OC); (4) DPDP-compliance and child-data-safety evaluation given [WEB-11, WEB-13]; and (5) multi-turn / learning-gains evaluation to address MRBench's own acknowledged limits [Q117, Q118].

ID	Evidence
Q1	"We propose a unified evaluation taxonomy with eight pedagogical dimensions based on key learning sciences principles, which is designed to assess the pedagogical value of LLM-powered AI tutor responses grounded in student mistakes or confusions in the mathematical domain." (p.1)
Q4	"General domain-agnostic natural language generation (NLG) metrics are not well-suited for this context, as most of them fail to account for pedagogical values and require gold references, which are often not available, especially in online interactions." (p.1)
Q15	"In this section, we first present our approach, narrowing the evaluation taxonomy down to eight measurable dimensions aligned with key pedagogical strategies (Jurenka et al., 2024; Hennessy et al., 2016). These dimensions are most suitable for the student mistake remediation task and are based on the learning sciences principles." (p.3)
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)
Q31	"We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient." (p.5)
Q38	"To measure inter-annotator agreement, we computed Fleiss' kappa value, which for this annotation experiment equals 0.65, indicating substantial agreement." (p.5)
Q57	"Each dimension was annotated using a three-tier labeling system (see Figure 1 and Table 4)." (p.6)
Q60	"The annotators reached an average Cohen's kappa score of 0.71, which indicates substantial inter-annotator agreement (McHugh, 2012)." (p.6)
Q63	"We utilize two key metrics to quantitatively assess the pedagogical effectiveness of LLMs and for comparative analysis: (1) Desired Annotation Match Rate (DAMR): This metric quantifies the percentage of responses from each human or LLM-based tutor that received the desired annotation labels." (p.6)

Q117	"However, one of the limitations is that it does not consider the tutoring dialogues' impact on the overall student learning." (p.9)
Q118	"The annotation pertains to the individual tutor turns within educational dialogues, which restricts our understanding of broader implications on student learning processes and learning gains, typically observed after a conversation concludes." (p.9)
Q145	"Table 8: Pedagogical ability assessment of different LLMs using the DAMR scores (in %) across eight evaluation dimensions with human evaluation on the Bridge data." (p.16)
Q146	"Table 9: Pedagogical ability assessment of different LLMs using the DAMR scores (in %) across eight evaluation dimensions with human evaluation on the MathDial data." (p.16)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench's output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-11	https://www.matters.ai/compliance/dpdp/dpdp-act-2023
WEB-13	71% of AI platforms used by minors in India are DPDP-non-compliant — regulatory context increases importance of Indian-stakeholder-validated labels for child-facing AI (https://asiaresearch.co/dpdp-compliance-in-respect-to-children-data/)

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: MRBENCH | Context: North India Grades 6–8 Mathematics Students (AI Tutoring)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MRBench defines a focused, well-motivated taxonomy for the mistake-remediation task in mathematics [Q1, Q12, Q14], which matches the deployment's core task at the task-family level. The eight dimensions are grounded in Western learning-sciences literature [Q15, Q16, Q17, Q18, Q19] and the authors explicitly acknowledge that the taxonomy would 'need to be verified on and likely adapted' for other tasks/subjects [Q115, Q127]. For the North India deployment, the taxonomy omits categories that would likely matter — culturally appropriate authority framing, code-switching handling, and bilingual scaffolding — because the annotators found no additional dimensions [Q39], but this reflects a homogeneous CS-postgraduate annotator pool [Q30, Q31] rather than an Indian teacher/student perspective. The 'not revealing the answer' norm [Q22] may itself be culturally dissonant in authority-oriented North Indian classroom contexts [cultural_norms_notes]. The category structure is thus partially aligned (mistake remediation is a real task type in the deployment) but underrepresents constructs relevant to bilingual, authority-oriented, NCERT-curriculum learning contexts.

ID	Evidence
Q1	"We propose a unified evaluation taxonomy with eight pedagogical dimensions based on key learning sciences principles, which is designed to assess the pedagogical value of LLM-powered AI tutor responses grounded in student mistakes or confusions in the mathematical domain." (p.1)
Q12	"In this work, we focus on educational dialogues between a student and a tutor in the mathematical domain. Specifically, the conversations are grounded in students' mistakes or confusions, and the AI tutor aims to respond in order to remediate such mistakes or confusions." (p.3)
Q14	"The evaluation taxonomy detailed in Section 4 assesses the appropriateness of the Tt+1 response across eight key pedagogical dimensions." (p.3)
Q15	"In this section, we first present our approach, narrowing the evaluation taxonomy down to eight measurable dimensions aligned with key pedagogical strategies (Jurenka et al., 2024; Hennessy et al., 2016). These dimensions are most suitable for the student mistake remediation task and are based on the learning sciences principles." (p.3)
Q16	"Encourage active learning (Chi and Wylie, 2014; Oakley and Sejnowski, 2021): The tutor should encourage students to actively participate in the discussion and practice rather than passively receive information. The tutor can achieve this by not revealing the answer immediately and scaffolding guidance." (p.3)
Q17	"Adapt to students' goals and needs (King and South, 2017): The tutor should respond coherently by adapting to the current state and goals of the student's learning rather than following a pre-defined learning path. In the context of student mistake remediation, this happens when the tutor identifies the mistake, pinpoints its location, and responds coherently." (p.3)
Q18	"Manage cognitive load and enhance metacognitive skills (Mayer, 2002; Dehaene, 2020; Cohen et al., 2021): The tutor should present the information in a structured manner, with elaboration and examples in manageably small chunks that enable the student to generalize their learning skills beyond the current problem. For the task at hand, this can be achieved by providing appropriate guidance." (p.3)
Q19	"Foster motivation and stimulate curiosity (Keller, 1987; Patall et al., 2008): The tutor" (p.3)
Q22	"Since most dialogues are relatively short and present contexts for the mistakes made early in the student's solution, a good tutor strategy is not to reveal the answer to the student immediately but rather provide helpful guidance." (p.4)
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)
Q31	"We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient." (p.5)
Q39	"None of the annotators identified any additional or redundant dimensions necessary for student mistake remediation." (p.5)

Q115	"We acknowledge that the proposed taxonomy will need to be verified on and likely adapted if applied to other tasks such as concept learning, and to subjects other than mathematics." (p.9)
Q127	"However, other educational settings, particularly those involving tutorial dialogues beyond mistake remediation, may require modifications, as discussed in the limitations section." (p.12)

Strengths

- The taxonomy is task-appropriate: it targets exactly the mistake-remediation setting the deployment envisions [Q1, Q12].
- It decomposes pedagogical quality into eight learning-sciences-grounded dimensions [Q15, Q16–Q19], giving fine-grained diagnostic categories rather than a single overall-quality label.
- The authors are explicit about scope limits and adaptation needs for other tasks/subjects [Q114, Q115, Q127], which supports principled porting.

ID	Evidence
Q1	"We propose a unified evaluation taxonomy with eight pedagogical dimensions based on key learning sciences principles, which is designed to assess the pedagogical value of LLM-powered AI tutor responses grounded in student mistakes or confusions in the mathematical domain." (p.1)
Q12	"In this work, we focus on educational dialogues between a student and a tutor in the mathematical domain. Specifically, the conversations are grounded in students' mistakes or confusions, and the AI tutor aims to respond in order to remediate such mistakes or confusions." (p.3)
Q15	"In this section, we first present our approach, narrowing the evaluation taxonomy down to eight measurable dimensions aligned with key pedagogical strategies (Jurenka et al., 2024; Hennessy et al., 2016). These dimensions are most suitable for the student mistake remediation task and are based on the learning sciences principles." (p.3)
Q16	"Encourage active learning (Chi and Wylie, 2014; Oakley and Sejnowski, 2021): The tutor should encourage students to actively participate in the discussion and practice rather than passively receive information. The tutor can achieve this by not revealing the answer immediately and scaffolding guidance." (p.3)
Q17	"Adapt to students' goals and needs (King and South, 2017): The tutor should respond coherently by adapting to the current state and goals of the student's learning rather than following a pre-defined learning path. In the context of student mistake remediation, this happens when the tutor identifies the mistake, pinpoints its location, and responds coherently." (p.3)
Q18	"Manage cognitive load and enhance metacognitive skills (Mayer, 2002; Dehaene, 2020; Cohen et al., 2021): The tutor should present the information in a structured manner, with elaboration and examples in manageably small chunks that enable the student to generalize their learning skills beyond the current problem. For the task at hand, this can be achieved by providing appropriate guidance." (p.3)
Q19	"Foster motivation and stimulate curiosity (Keller, 1987; Patall et al., 2008): The tutor" (p.3)
Q114	"The proposed taxonomy primarily focuses on the task of the student mistake remediation in the domain of mathematics." (p.9)
Q115	"We acknowledge that the proposed taxonomy will need to be verified on and likely adapted if applied to other tasks such as concept learning, and to subjects other than mathematics." (p.9)
Q127	"However, other educational settings, particularly those involving tutorial dialogues beyond mistake remediation, may require modifications, as discussed in the limitations section." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: For North India Grades 6–8 AI tutoring, required categories include: mistake remediation on NCERT/CBSE curriculum topics (fractions, integers, linear equations, mensuration, Indian numeral system) [curriculum_context], handling of Hinglish/code-switched student input [WEB-1], culturally appropriate authority/tone framing [cultural_norms_notes], and factual-correctness safeguards for a population with limited independent verification access [domain_specific_notes.factual_accuracy_risk].

IO-2: The MRBench taxonomy omits categories relevant to the region: it has no dimension for code-switching handling or language-register appropriateness [Q51 — text-only, English-only setup; WEB-1], no dimension for cultural/authority appropriateness of tutor stance [Q22, Q26; cultural_norms_notes], and no explicit dimension for curriculum/notation alignment (e.g., Indian numeral system, rupee word problems) [WEB-7, WEB-8]. Annotators identified no additional dimensions [Q39], but the annotator pool [Q30, Q31] would not have surfaced these.

IO-3: Most included categories (mistake identification, guidance, actionability, coherence, revealing-the-answer, human-likeness) remain relevant. The 'not revealing the answer' criterion [Q22] and constructivist 'encourage active learning' framing [Q16] may be partially misaligned with authority-oriented Indian classroom expectations, but they are not irrelevant — NEP 2020 aspires toward inquiry-based learning [WEB-10, WEB-9], so these categories are contested rather than inapplicable.

IO-4: Gaps that would harm content validity: (a) absence of a code-switching / language-appropriateness category [WEB-1]; (b) absence of a curriculum/notation-fit category [WEB-7, WEB-8]; (c) absence of an authority/tone-fit category calibrated to North Indian norms [cultural_norms_notes]. The tone dimension collapses to 'non-offensive' [Q88], further indicating limited discriminative power for culturally specific tone norms.

ID	Evidence
Q16	"Encourage active learning (Chi and Wylie, 2014; Oakley and Sejnowski, 2021): The tutor should encourage students to actively participate in the discussion and practice rather than passively receive information. The tutor can achieve this by not revealing the answer immediately and scaffolding guidance." (p.3)
Q22	"Since most dialogues are relatively short and present contexts for the mistakes made early in the student's solution, a good tutor strategy is not to reveal the answer to the student immediately but rather provide helpful guidance." (p.4)
Q26	"In addition to addressing student mistakes, a good tutor should encourage them and avoid using toxic language, which is aligned with the care dimension in the evaluation schema of Wang et al. (2024a)." (p.4)
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)
Q31	"We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient." (p.5)
Q39	"None of the annotators identified any additional or redundant dimensions necessary for student mistake remediation." (p.5)
Q51	"For the 192 instances in MRBench (60 from Bridge and 132 from MathDial), we generated appropriate subsequent responses based on the conversation history and the last utterance, which contained confusions or mistakes, using seven state-of-the-art LLMs." (p.6)
Q88	"When we combine these two labels into "Non-offensive", the DAMR score reaches 100% as we observe no offensive responses from any LLMs or humans." (p.8)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench's output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-7	NCERT Class 6 numeral conventions differ from MRBench; annotators would need Indian curricular familiarity to judge notation-related responses (https://edusaksham.com/chapters/CBSE-Class-6-Mathematics-Numeration-Systems-Notes.html)
WEB-8	NCERT Class 7 Ganita Prakash confirms lakh/crore use in problems (https://edurev.in/t/407589/ncert-solutions-class-7-mathematics-ganita-prakash-large-numbers-around-us)
WEB-9	https://www.orfonline.org/expert-speak/five-years-of-nep-2020-and-the-promise-of-edtech
WEB-10	NEP 2020 promotes inquiry-based learning aspirationally (https://static.pib.gov.in/WriteReadData/userfiles/NEP_Final_English_0.pdf)

Information Gaps

- Whether Indian teacher/student judgment would introduce additional taxonomy dimensions has not been empirically tested.

- Whether the current eight-dimension set covers culturally-specific pedagogical constructs (e.g., guru-shishya authority dynamics) is unverified.

Requires Expert Verification

- Whether North Indian teachers would judge 'not revealing the answer' as a valid pedagogical criterion in Grades 6–8 mathematics contexts.
- Which additional pedagogical dimensions Indian educators would insist on.

Remediation

Gap 1: Taxonomy lacks categories for code-switching handling, curriculum/notation fit, and culturally calibrated authority/tone — none surfaced by the current annotator pool [Q39, Q30, Q31].

Recommendation: Extend the taxonomy with three additional dimensions — 'language-register appropriateness', 'curriculum/notation fit', and 'cultural tone/authority calibration' — and pilot them with Indian annotators to establish inter-annotator agreement before large-scale use.

ID	Evidence
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)
Q31	"We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient." (p.5)
Q39	"None of the annotators identified any additional or redundant dimensions necessary for student mistake remediation." (p.5)

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MRBENCH | Context: North India Grades 6–8 Mathematics Students (AI Tutoring)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MRBench inputs are drawn entirely from two Western-origin datasets — Bridge (US elementary tutoring dialogues) [Q40, Q43, Q45] and MathDial (US-derived middle-school reasoning) [Q47, Q48] — with no documented connection to Indian curriculum, notation, cultural context, or linguistic register. The deployment targets NCERT/CBSE Grades 6–8 students, where the Indian numeral system (lakh, crore, Indian comma grouping) is explicitly taught from Class 6 and word problems routinely use rupees and Indian names/settings [WEB-7, WEB-8]. None of these appear in MRBench. Additionally, the target population is bilingual, with common Hindi–English code-switching [WEB-1, WEB-19], and MRBench contains no Hindi, Devanagari, transliterated Hindi, Hinglish, or Indian-English register instances. This is a systematic content-validity failure: the concrete datapoints reflect a culturally, linguistically, and curricularly foreign distribution.

ID	Evidence
Q40	"We have compiled mistake remediation benchmark, MRBench, from the Bridge (Wang et al., 2024a) and MathDial (Macina et al., 2023) datasets." (p.5)
Q43	"The Bridge dataset (Wang et al., 2024a) comprises partial dialogue interactions between real human tutors and students at the elementary level, featuring two distinct human tutor responses (novice and expert)." (p.5)
Q45	"The original dataset consists of a total of 700 dialogues; we filtered 60 high-quality instances for MRBench." (p.5)
Q47	"The dialogues in the MathDial dataset (Macina et al., 2023) consist of complete multi-turn conversations between a real human tutor and an LLM acting as a student, where the tutor aims to remediate the student's mistakes." (p.5)
Q48	"Specifically, these conversations are grounded in middle school-level mathematical reasoning questions." (p.5)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench's output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-7	NCERT Class 6 numeral conventions differ from MRBench; annotators would need Indian curricular familiarity to judge notation-related responses (https://edusaksham.com/chapters/CBSE-Class-6-Mathematics-Numeration-Systems-Notes.html)
WEB-8	NCERT Class 7 Ganita Prakash confirms lakh/crore use in problems (https://edurev.in/t/407589/ncert-solutions-class-7-mathematics-ganita-prakash-large-numbers-around-us)
WEB-19	HiACC corpus documents Hindi–English code-switching in children's communication contexts (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12329218/)

Strengths

- The source dialogues are anchored in genuine student mistakes at levels overlapping the target grade band (elementary via Bridge; middle school via MathDial) [Q43, Q48], so the task grounding (mistakes in arithmetic and reasoning) is domain-appropriate even if surface content differs.
- Manual filtering was applied to both source datasets to retain high-quality instances [Q45, Q46, Q50], reducing noise in the underlying task.

ID	Evidence
Q43	"The Bridge dataset (Wang et al., 2024a) comprises partial dialogue interactions between real human tutors and students at the elementary level, featuring two distinct human tutor responses (novice and expert)." (p.5)
Q45	"The original dataset consists of a total of 700 dialogues; we filtered 60 high-quality instances for MRBench." (p.5)

Q46	“Among the various criteria for selecting high-quality dialogues, the key one was that the student’s last utterance (or last few utterances) should exhibit an error or confusion.” (p.5)
Q48	“Specifically, these conversations are grounded in middle school-level mathematical reasoning questions.” (p.5)
Q50	“To further ensure the reliability of our benchmark, we manually inspected the data in order to retain only high-quality examples, which resulted in 132 instances for MRBench.” (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — authentic deployment input in North India Grades 6–8 would include Indian numeral conventions (lakh/crore) [WEB-7, WEB-8], rupee-denominated word problems [WEB-7], Indian names and cultural settings, and Hinglish/code-switched phrasing [WEB-1, WEB-19]. Region-specific knowledge is intrinsic to authentic student utterances.

IC-2: MRBench content is drawn from Bridge and MathDial [Q40, Q43, Q47], both Western-origin. There is no documented alignment with Indian cultural context; the content does not align with the target deployment culture.

IC-3: Bridge uses US elementary arithmetic dialogues [Q43, Q44]; MathDial uses US-style middle-school reasoning problems [Q47, Q48, Q92]. These carry Western pedagogical framing, notation, and word-problem conventions that may not transfer to NCERT/CBSE learners [WEB-7, WEB-8].

IC-4: Not done in MRBench — the annotator pool has no stated Indian educational connection [Q30, Q31]. Recruiting NCERT-familiar Indian annotators to flag culturally sensitive or curriculum-mismatched instances would be necessary.

IC-5: Content-validity issues: (a) systematic curriculum/notation mismatch (Indian numeral system, rupees) absent from MRBench [WEB-7, WEB-8]; (b) absence of Indian English register and Hinglish [Q51; WEB-1]; (c) Western word-problem cultural framing (names, settings) throughout. These constitute construct-irrelevant variance for the target deployment.

ID	Evidence
Q30	“The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English.” (p.5)
Q31	“We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient.” (p.5)
Q40	“We have compiled mistake remediation benchmark, MRBench, from the Bridge (Wang et al., 2024a) and MathDial (Macina et al., 2023) datasets.” (p.5)
Q43	“The Bridge dataset (Wang et al., 2024a) comprises partial dialogue interactions between real human tutors and students at the elementary level, featuring two distinct human tutor responses (novice and expert).” (p.5)
Q44	“The dialogue context is typically short (few turns) and predominantly focused on fundamental mathematical concepts, including operations such as multiplication, addition, and so on.” (p.5)
Q47	“The dialogues in the MathDial dataset (Macina et al., 2023) consist of complete multi-turn conversations between a real human tutor and an LLM acting as a student, where the tutor aims to remediate the student’s mistakes.” (p.5)
Q48	“Specifically, these conversations are grounded in middle school-level mathematical reasoning questions.” (p.5)
Q51	“For the 192 instances in MRBench (60 from Bridge and 132 from MathDial), we generated appropriate subsequent responses based on the conversation history and the last utterance, which contained confusions or mistakes, using seven state-of-the-art LLMs.” (p.6)

Q92	"In contrast, for MathDial, the contexts are longer, the mistakes are grounded in reasoning, and the responses are more structured." (p.8)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench's output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-7	NCERT Class 6 numeral conventions differ from MRBench; annotators would need Indian curricular familiarity to judge notation-related responses (https://edusaksham.com/chapters/CBSE-Class-6-Mathematics-Numeration-Systems-Notes.html)
WEB-8	NCERT Class 7 Ganita Prakash confirms lakh/crore use in problems (https://edurev.in/t/407589/ncert-solutions-class-7-mathematics-ganita-prakash-large-numbers-around-us)
WEB-19	HiACC corpus documents Hindi–English code-switching in children's communication contexts (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12329218/)

Information Gaps

- Exact overlap between MathDial problem topics and NCERT Grades 6–8 syllabus has not been quantified.
- No dataset-level analysis of MRBench instances was available to confirm the absence of Indian content empirically (documentation-based inference only).

Requires Expert Verification

- Whether any MathDial problems happen to align with NCERT topics in ways that permit partial reuse without adaptation.

Remediation

Gap 1: MRBench contains no NCERT/CBSE-aligned problems, no Indian numeral conventions (lakh/crore), no rupee word problems, and no Indian English / Hinglish / Devanagari inputs [Q40, Q43, Q47; WEB-7, WEB-8, WEB-1].

Recommendation: Construct a supplementary evaluation set of at least ~200 mistake-remediation dialogues sampled from NCERT Grades 6–8 mathematics topics (integers, fractions, linear equations, mensuration), including Indian-numeral and rupee word problems, with student utterances drawn from real Indian classroom interactions where feasible; include Hinglish and Romanized-Hindi variants to test code-switching handling.

ID	Evidence
Q40	"We have compiled mistake remediation benchmark, MRBench, from the Bridge (Wang et al., 2024a) and MathDial (Macina et al., 2023) datasets." (p.5)
Q43	"The Bridge dataset (Wang et al., 2024a) comprises partial dialogue interactions between real human tutors and students at the elementary level, featuring two distinct human tutor responses (novice and expert)." (p.5)
Q47	"The dialogues in the MathDial dataset (Macina et al., 2023) consist of complete multi-turn conversations between a real human tutor and an LLM acting as a student, where the tutor aims to remediate the student's mistakes." (p.5)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench's output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-7	NCERT Class 6 numeral conventions differ from MRBench; annotators would need Indian curricular familiarity to judge notation-related responses (https://edusaksham.com/chapters/CBSE-Class-6-Mathematics-Numeration-Systems-Notes.html)
WEB-8	NCERT Class 7 Ganita Prakash confirms lakh/crore use in problems (https://edurev.in/t/407589/ncert-solutions-class-7-mathematics-ganita-prakash-large-numbers-around-us)

Input Form (IF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MRBENCH | Context: North India Grades 6–8 Mathematics Students (AI Tutoring)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MRBench is text-only, English-only, and single-modality [Q51, Q53]. Input instances are conversation histories terminated at a student mistake [Q49, Q51], with lengths and turn counts varying across Bridge and MathDial [Q138, Q139, Q143]. For the North India deployment, this form choice omits three signals central to authentic input: (1) Hinglish and Romanized-Hindi input [WEB-1, WEB-19]; (2) Devanagari script or mixed-script input [writing_systems.digital_input_note, WEB-1]; and (3) audio/voice input relevant to lower-literacy or lower-English-proficiency segments [infrastructure_notes]. Since Indi-RomCoM shows that LLM performance degrades on Romanized code-mixed inputs [WEB-1], evaluating tutors on English-only MRBench likely overestimates deployment performance — an external-validity violation. English text is workable at the surface level for the more English-proficient urban private-school subset, but not for the broader target population.

ID	Evidence
Q49	“To match the format of Bridge (partial conversations with the last few student’s utterances exhibiting a mistake or confusion), we prepared the dataset by terminating a conversation where the student makes a mistake and considering the next tutor response as the expert tutor response (there are no associated novice responses).” (p.5)
Q51	“For the 192 instances in MRBench (60 from Bridge and 132 from MathDial), we generated appropriate subsequent responses based on the conversation history and the last utterance, which contained confusions or mistakes, using seven state-of-the-art LLMs.” (p.6)
Q53	“Each LLM has associated responses for 192 dialogues, resulting in a benchmark of 192×7 (7 LLM responses) + 192×1 (expert responses) + 60×1 (novice responses) = 1,596 responses, which makes the evaluation benchmark reasonably large while still manageable for human annotation described in Section 5.2.” (p.6)
Q138	“It can be observed that the conversation history and response lengths from different LLMs and humans are generally shorter in the Bridge dataset compared to the MathDial dataset.” (p.14)
Q139	“Additionally, the number of turns differs between them.” (p.14)
Q143	“In all cases, length is estimated using the number of characters.” (p.14)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench’s output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-19	HiACC corpus documents Hindi–English code-switching in children’s communication contexts (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12329218/)

Strengths

- Text-only format is aligned with the dominant text-input modality of chatbot-style AI tutors on smartphones, which is the likely device profile for the deployment [infrastructure_notes].
- The paper documents length and turn-count differences between Bridge and MathDial [Q138, Q139, Q143], enabling downstream analysis of form-related performance variation.

ID	Evidence
Q138	“It can be observed that the conversation history and response lengths from different LLMs and humans are generally shorter in the Bridge dataset compared to the MathDial dataset.” (p.14)
Q139	“Additionally, the number of turns differs between them.” (p.14)

Q143	"In all cases, length is estimated using the number of characters." (p.14)
------	--

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal-distribution mismatch: MRBench inputs are English-only text [Q51]; deployment inputs will include Indian English, Hinglish, Romanized Hindi, and potentially Devanagari [WEB-1, WEB-19; writing_systems]. Indi-RomCoM empirically shows LLM degradation on Romanized code-mixed inputs [WEB-1], indicating the mismatch has performance consequences.

IF-2: Regional infrastructure supports text input on smartphones (>88% household smartphone access for teens per ASER 2024 [WEB-5]), but the dominant input mode is Roman-script transliterated Hindi rather than Devanagari [writing_systems.digital_input_note], which MRBench does not cover. Voice input is important for lower-literacy segments but MRBench is text-only.

IF-3: Domain-specific form differences: (a) no code-switched inputs [Q51; WEB-1]; (b) no script-mixed inputs [WEB-1]; (c) no audio/voice inputs [Q51], relevant for lower-English-proficiency learners in this deployment. The prompt template used to elicit LLM responses is adapted from a Western study [Q131], embedding further Western-form assumptions.

IF-4: External-validity mismatches: (a) English-only text underrepresents authentic bilingual student input [WEB-1]; (b) absence of transliterated/Devanagari input means the benchmark cannot probe script-handling capabilities relevant to the deployment [writing_systems]; (c) absence of audio limits accessibility evaluation [infrastructure_notes].

ID	Evidence
Q51	"For the 192 instances in MRBench (60 from Bridge and 132 from MathDial), we generated appropriate subsequent responses based on the conversation history and the last utterance, which contained confusions or mistakes, using seven state-of-the-art LLMs." (p.6)
Q131	"The template is adapted from Wang et al. (2024a)." (p.12)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench's output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-5	ASER 2024: ~57% of teens with household smartphones use them for study — some deployments may rely on voice/audio output not evaluated here (https://thefederal.com/category/education/schools/9-in-10-indian-teens-have-a-smartphone-not-even-6-in-10-use-it-to-learn-242233)
WEB-6	PIB Telecom Survey 2025: rural smartphone ownership among mobile users ~95.5% in 15–29 age band (https://www.pib.gov.in/PressReleasePage.aspx?PRID=2132330)
WEB-19	HiACC corpus documents Hindi–English code-switching in children's communication contexts (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12329218/)

Information Gaps

- The exact prevalence of Hinglish/Romanized-Hindi input by North Indian Grade 6–8 students in AI-tutor interactions has not been directly measured.
- Whether the deployment envisions voice input has not been specified.

Requires Expert Verification

- Prevalence of code-switching in actual student typing behavior in this cohort.

Remediation

Gap 1: English-only text input does not cover Hinglish, Romanized-Hindi, Devanagari, or mixed-script inputs common in the target population [Q51; WEB-1, WEB-19].

Recommendation: Add a code-mixed input track to any MRBench-based evaluation, using Indi-RomCoM-style [WEB-1] intensity levels of code-mixing; report DAMR separately for English-only vs. code-mixed conditions to expose deployment-relevant degradation.

ID	Evidence
Q51	“For the 192 instances in MRBench (60 from Bridge and 132 from MathDial), we generated appropriate subsequent responses based on the conversation history and the last utterance, which contained confusions or mistakes, using seven state-of-the-art LLMs.” (p.6)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench’s output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-19	HiACC corpus documents Hindi–English code-switching in children’s communication contexts (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12329218/)

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: MRBENCH | Context: North India Grades 6–8 Mathematics Students (AI Tutoring)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output ontology uses a three-tier categorical labeling per dimension with 'desired' labels specified [Q57, Q58, Q64, Q125]. Its decision rules encode normative Western pedagogical positions — e.g., 'not revealing the answer' is a desired behavior [Q22]; guidance must be factually correct to count [Q82]; tutor tone should be non-offensive [Q88]. Two concrete signals of misalignment for the target deployment: (1) the tone dimension collapses to 100% non-offensive across all tutors when 'encouraging' and 'neutral' are merged [Q88], indicating the label structure lacks the resolution to distinguish culturally calibrated encouragement (which North Indian norms treat as more restrained [cultural_norms_notes]); (2) the 'not revealing the answer' desired label [Q22] encodes a constructivist norm that is potentially at odds with authority-oriented classroom expectations described in cultural_norms_notes, though NEP 2020 [WEB-10] moves aspirationally in this direction. Desired labels were not validated against Indian teacher/student judgment [Q30, Q31]. Structural and content validity are therefore compromised.

ID	Evidence
Q22	"Since most dialogues are relatively short and present contexts for the mistakes made early in the student's solution, a good tutor strategy is not to reveal the answer to the student immediately but rather provide helpful guidance." (p.4)
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)
Q31	"We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient." (p.5)
Q57	"Each dimension was annotated using a three-tier labeling system (see Figure 1 and Table 4)." (p.6)
Q58	"For instance, the 'mistake identification' dimension employed the following labels: (i) yes, (ii) to some extent, and (iii) no." (p.6)
Q64	"The desired labels for each dimension are detailed in Table 2." (p.6)
Q82	"Can a tutor achieve a higher DAMR score for actionability while receiving a lower score for providing guidance? This is possible since we consider only factually correct guidance as useful (see Table 4)." (p.8)
Q88	"When we combine these two labels into 'Non-offensive', the DAMR score reaches 100% as we observe no offensive responses from any LLMs or humans." (p.8)
Q125	"The definitions, associated labels, and the desired labels for each dimension of the proposed taxonomy are provided in Table 4." (p.12)
WEB-10	NEP 2020 promotes inquiry-based learning aspirationally (https://static.pib.gov.in/WriteReadData/userfiles/NEP_Final_English_0.pdf)

Strengths

- The three-tier per-dimension labeling with explicit desired labels [Q57, Q64, Q125] gives an interpretable, dimension-independent scoring space.
- The paper explicitly acknowledges interdependencies among dimensions but justifies treating them as independent for annotation [Q29], and separates guidance (factual correctness) from actionability [Q82, Q83], which surfaces meaningful trade-offs.

- The authors are transparent about limits (e.g., dimensions may be interrelated in practice [Q113]).

ID	Evidence
Q29	"Although there are inherent interdependencies among the proposed dimensions of the taxonomy (e.g., a response that reveals the answer is less likely to be actionable, and vice versa), we explicitly instructed all annotators to treat each dimension as independent and orthogonal to minimize confounding factors and potential biases during the annotation process." (p.4)
Q57	"Each dimension was annotated using a three-tier labeling system (see Figure 1 and Table 4)." (p.6)
Q64	"The desired labels for each dimension are detailed in Table 2." (p.6)
Q82	"Can a tutor achieve a higher DAMR score for actionability while receiving a lower score for providing guidance? This is possible since we consider only factually correct guidance as useful (see Table 4)." (p.8)
Q83	"At the same time, even incorrect or incomplete guidance can lead to certain actions on the part of the student and can foster their curiosity, thus providing them with learning opportunities." (p.8)
Q113	"However, in practice, these dimensions may be inherently interrelated and may influence one another." (p.9)
Q125	"The definitions, associated labels, and the desired labels for each dimension of the proposed taxonomy are provided in Table 4." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Categories are mostly relevant at the abstract level (mistake identification, guidance, actionability, coherence, tone, human-likeness) [Q57, Q125]. However, their desired labels encode Western pedagogical defaults [Q22, Q82, Q88] that may not match regional expectations.

OO-2: Missing categories relevant to the region include: language-appropriateness / code-switching handling [WEB-1], cultural authority-tone fit [cultural_norms_notes], curriculum-notation fit (Indian numeral system, rupees) [WEB-7, WEB-8], and hallucination-risk / factual-safety category calibrated to student vulnerability [Q121, Q122; domain_specific_notes.factual_accuracy_risk].

OO-3: 'Not revealing the answer' as a desired label [Q22] encodes a constructivist norm that may conflict with authority-oriented classroom expectations [cultural_norms_notes]. The tone dimension's 'encouraging vs. neutral' distinction collapses in practice [Q88], suggesting the encoded label structure does not capture culturally specific encouragement norms.

OO-4: Stakeholder-driven redesign is needed: Indian teachers and student representatives should validate desired labels, especially for tone, revealing-the-answer, and motivation/encouragement dimensions.

OO-5: Documentation issues that harm structural and content validity: (a) desired labels not validated against Indian stakeholders [Q30, Q31]; (b) tone-label collapse [Q88] indicates insufficient resolution; (c) no output category tracks language-register or code-switching appropriateness [Q51, WEB-1].

ID	Evidence
Q22	"Since most dialogues are relatively short and present contexts for the mistakes made early in the student's solution, a good tutor strategy is not to reveal the answer to the student immediately but rather provide helpful guidance." (p.4)
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)

Q31	"We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient." (p.5)
Q51	"For the 192 instances in MRBench (60 from Bridge and 132 from MathDial), we generated appropriate subsequent responses based on the conversation history and the last utterance, which contained confusions or mistakes, using seven state-of-the-art LLMs." (p.6)
Q57	"Each dimension was annotated using a three-tier labeling system (see Figure 1 and Table 4)." (p.6)
Q82	"Can a tutor achieve a higher DAMR score for actionability while receiving a lower score for providing guidance? This is possible since we consider only factually correct guidance as useful (see Table 4)." (p.8)
Q88	"When we combine these two labels into 'Non-offensive', the DAMR score reaches 100% as we observe no offensive responses from any LLMs or humans." (p.8)
Q121	"Although we do not foresee any ethical risks, we acknowledge that this work relies on the outputs from LLMs, and there are certain risks associated with such outputs in general since these models may generate responses that, although plausible, can be factually incorrect, nonsensical, or even offensive." (p.9)
Q122	"Of particular importance for the educational domain is the fact that hallucinations can misguide students and propagate biases." (p.9)
Q125	"The definitions, associated labels, and the desired labels for each dimension of the proposed taxonomy are provided in Table 4." (p.12)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench's output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-7	NCERT Class 6 numeral conventions differ from MRBench; annotators would need Indian curricular familiarity to judge notation-related responses (https://edusaksham.com/chapters/CBSE-Class-6-Mathematics-Numeration-Systems-Notes.html)
WEB-8	NCERT Class 7 Ganita Prakash confirms lakh/crore use in problems (https://edurev.in/t/407589/ncert-solutions-class-7-mathematics-ganita-prakash-large-numbers-around-us)

Information Gaps

- No empirical comparison of desired-label distributions between the MRBench annotator pool and Indian teachers/students exists.
- The distinction between aspirational NEP norms and actual North Indian classroom norms may or may not affect stakeholder-desired labels.

Requires Expert Verification

- Whether Indian teachers would endorse 'not revealing the answer' as the desired label in Grades 6–8 mistake-remediation contexts.
- Whether Indian stakeholders would endorse the current tone/encouragement label structure or require culturally-calibrated alternatives.

Remediation

Gap 1: Desired labels encode Western pedagogical defaults (e.g., 'not revealing the answer' [Q22]) not validated against Indian stakeholders; tone label structure collapses in practice [Q88].

Recommendation: Convene an Indian teacher-and-student advisory panel to review desired labels for each of the eight dimensions and to propose culturally-calibrated tone/encouragement labels; extend the ontology with a language-appropriateness/code-switching dimension and a curriculum/notation-fit dimension.

ID	Evidence
Q22	"Since most dialogues are relatively short and present contexts for the mistakes made early in the student's solution, a good tutor strategy is not to reveal the answer to the student immediately but rather provide helpful guidance." (p.4)
Q88	"When we combine these two labels into 'Non-offensive', the DAMR score reaches 100% as we observe no offensive responses from any LLMs or humans." (p.8)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MRBENCH | Context: North India Grades 6–8 Mathematics Students (AI Tutoring)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Ground-truth annotations come from four in-house annotators — CS postgraduates, English-proficient, with no required teaching experience [Q30, Q31] — recruited and trained rigorously [Q32, Q33, Q34, Q35, Q36, Q37], with strong internal agreement (Fleiss' κ 0.65 pilot [Q38], Cohen's κ 0.71 main [Q60]). Human evaluations are treated as gold standard [Q68]. For the target deployment, this pool is homogeneous and unrepresentative: no stated connection to Indian education, NCERT curriculum, Hindi–English bilingualism, or Grades 6–8 teaching practice. The explicit choice to view annotators as 'potential users of AI tutors (or a student)' rather than teachers [Q31] further distances labels from the perspective of Indian classroom stakeholders. This threatens convergent validity (labels may not correlate with regional judgment) and external validity (findings may not generalize). LLM-annotator alternatives (Prometheus2, Llama-3.1-8B) performed poorly [Q102, Q104, Q106], so there is no ready fallback. Internal quality controls are a real strength, so the score is not the floor.

ID	Evidence
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)
Q31	"We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient." (p.5)
Q32	"To control the annotation workflow and ensure quality, we opted not to use public annotation outsourcing platforms such as Prolific or MTurk, which allowed us to implement rigorous training protocols and a robust validation mechanism for the annotations." (p.5)
Q33	"First, we provided all annotators with comprehensive training, including an interactive training document (see Section C for more details) and oral instructions." (p.5)
Q34	"Following this, we conducted validation pilot study to evaluate the annotation quality and the annotators' understanding of the instructions before rolling out the large-scale human evaluation detailed in Section 5.2." (p.5)
Q35	"In this validation pilot study, all four annotators iteratively reviewed the annotation scheme and guidelines." (p.5)
Q36	"Each annotator also independently labeled the same eight randomly sampled dialogues – four from each of the two datasets (Bridge and MathDial) – across the eight dimensions of the evaluation taxonomy." (p.5)
Q37	"Given that each dialogue contained multiple responses from both LLMs and humans, and each response was annotated across eight evaluation dimensions, this resulted in a total of 544 annotations per annotator." (p.5)
Q38	"To measure inter-annotator agreement, we computed Fleiss' kappa value, which for this annotation experiment equals 0.65, indicating substantial agreement." (p.5)
Q60	"The annotators reached an average Cohen's kappa score of 0.71, which indicates substantial inter-annotator agreement (McHugh, 2012)." (p.6)
Q68	"We consider human-based evaluations as gold standard." (p.7)
Q102	"We also performed annotations using Prometheus2 and Llama-3.1-8B as critic LLMs." (p.8)
Q104	"Across both LLMs, it can be observed that most of the correlation scores are negative (except for the human-likeness dimension), indicating that the annotations from the LLMs are unreliable for the challenging pedagogical dimensions." (p.8)
Q106	"We believe both LLMs have a limited understanding of rich pedagogical concepts, as they were not specifically trained on pedagogically rich datasets." (p.8)

Strengths

- Rigorous training and quality controls: comprehensive training document, pilot with iterative review, structured quiz [Q32, Q33, Q34, Q35, Q133, Q134].
- Substantial inter-annotator agreement — Fleiss' κ 0.65 (pilot) [Q38] and Cohen's κ 0.71 (main) [Q60].
- Transparent reporting of annotator background, gender balance, and language proficiency [Q30]; explicit rationale for not using crowdsourcing [Q32].
- Acknowledgment of ethical/hallucination risks specifically in educational contexts [Q121, Q122].

ID	Evidence
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)
Q32	"To control the annotation workflow and ensure quality, we opted not to use public annotation outsourcing platforms such as Prolific or MTurk, which allowed us to implement rigorous training protocols and a robust validation mechanism for the annotations." (p.5)
Q33	"First, we provided all annotators with comprehensive training, including an interactive training document (see Section C for more details) and oral instructions." (p.5)
Q34	"Following this, we conducted validation pilot study to evaluate the annotation quality and the annotators' understanding of the instructions before rolling out the large-scale human evaluation detailed in Section 5.2." (p.5)
Q35	"In this validation pilot study, all four annotators iteratively reviewed the annotation scheme and guidelines." (p.5)
Q38	"To measure inter-annotator agreement, we computed Fleiss' kappa value, which for this annotation experiment equals 0.65, indicating substantial agreement." (p.5)
Q60	"The annotators reached an average Cohen's kappa score of 0.71, which indicates substantial inter-annotator agreement (McHugh, 2012)." (p.6)
Q121	"Although we do not foresee any ethical risks, we acknowledge that this work relies on the outputs from LLMs, and there are certain risks associated with such outputs in general since these models may generate responses that, although plausible, can be factually incorrect, nonsensical, or even offensive." (p.9)
Q122	"Of particular importance for the educational domain is the fact that hallucinations can misguide students and propagate biases." (p.9)
Q133	"Subsequently, we assessed annotators' understanding through a structured quiz, as is shown in a screenshot presented in Figure 5." (p.13)
Q134	"Additionally, we developed a comprehensive set of annotation guidelines, serving as a reference for annotators during the large-scale annotation process." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Labels reflect the perspectives of CS-postgraduate annotators in Abu Dhabi [Q6, Q30], not North Indian teachers, parents, or Grades 6–8 students. Regional stakeholder perspectives are unrepresented.

OC-2: Disagreement is plausible on culturally loaded dimensions — tone/encouragement [Q88; cultural_norms_notes], revealing-the-answer [Q22; cultural_norms_notes], motivational framing [Q19] — and on curriculum/notation-related

judgments [WEB-7, WEB-8]. No empirical measure exists, but the compositional gap makes disagreement likely.

OC-3: Annotator demographics are partially documented: four annotators, two male and two female, ≥postgraduate CS, English-proficient [Q30], no teaching experience required [Q31]. Nationality, first language, and educational background beyond degree level are not reported.

OC-4: Re-annotation by an Indian teacher-and-student annotator pool is needed to establish convergent validity for the target deployment. Not done in MRBench.

OC-5: The paper does not discuss aggregation-related erasure of minority perspectives explicitly, but with only four annotators, minority views are collapsed via kappa-style agreement targets [Q38, Q60]. This is documentation-silent on which perspectives may be lost. INSUFFICIENT DOCUMENTATION on formal aggregation mechanics.

OC-6: Label-content issues: (a) annotator pool unrepresentative of target region [Q30, Q31]; (b) LLM-annotator fallback unreliable [Q102, Q104]; (c) no validation of labels against Indian teachers/students. These harm both convergent and external validity.

ID	Evidence
Q6	"Kaushal Kumar Maurya, KV Aditya Srivatsa, Kseniia Petukhova and Ekaterina Kochmar, Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, UAE" (p.1)
Q19	"Foster motivation and stimulate curiosity (Keller, 1987; Patall et al., 2008): The tutor" (p.3)
Q22	"Since most dialogues are relatively short and present contexts for the mistakes made early in the student's solution, a good tutor strategy is not to reveal the answer to the student immediately but rather provide helpful guidance." (p.4)
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)
Q31	"We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient." (p.5)
Q38	"To measure inter-annotator agreement, we computed Fleiss' kappa value, which for this annotation experiment equals 0.65, indicating substantial agreement." (p.5)
Q60	"The annotators reached an average Cohen's kappa score of 0.71, which indicates substantial inter-annotator agreement (McHugh, 2012)." (p.6)
Q88	"When we combine these two labels into "Non-offensive", the DAMR score reaches 100% as we observe no offensive responses from any LLMs or humans." (p.8)
Q102	"We also performed annotations using Prometheus2 and Llama-3.1-8B as critic LLMs." (p.8)
Q104	"Across both LLMs, it can be observed that most of the correlation scores are negative (except for the human-likeness dimension), indicating that the annotations from the LLMs are unreliable for the challenging pedagogical dimensions." (p.8)
WEB-7	NCERT Class 6 numeral conventions differ from MRBench; annotators would need Indian curricular familiarity to judge notation-related responses (https://edusaksham.com/chapters/CBSE-Class-6-Mathematics-Numeration-Systems-Notes.html)
WEB-8	NCERT Class 7 Ganita Prakash confirms lakh/crore use in problems (https://edurev.in/t/407589/ncert-solutions-class-7-mathematics-ganita-prakash-large-numbers-around-us)

Information Gaps

- Annotator nationality, first language, and cultural background beyond 'proficient in English' and CS-postgraduate status are not disclosed [Q30].
- No empirical comparison of MRBench labels to Indian-stakeholder judgments exists.

Requires Expert Verification

- Extent of label disagreement between the MRBench annotator pool and North Indian teachers/students on the eight dimensions.

Remediation

Gap 1: Annotator pool is four CS-postgraduates with no teaching experience and no stated Indian educational context [Q30, Q31]; LLM-annotator fallback is unreliable [Q102, Q104].

Recommendation: Re-annotate a stratified MRBench subset with an Indian annotator pool that includes practicing Grades 6–8 mathematics teachers and bilingual Indian students, and quantify disagreement with the original labels dimension-by-dimension to test convergent validity.

ID	Evidence
Q30	"The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English." (p.5)
Q31	"We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient." (p.5)
Q102	"We also performed annotations using Prometheus2 and Llama-3.1-8B as critic LLMs." (p.8)
Q104	"Across both LLMs, it can be observed that most of the correlation scores are negative (except for the human-likeness dimension), indicating that the annotations from the LLMs are unreliable for the challenging pedagogical dimensions." (p.8)

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: MRBENCH | Context: North India Grades 6–8 Mathematics Students (AI Tutoring)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The evaluated output form is single-turn English text scored via categorical DAMR [Q63, Q64, Q65] and Pearson-based AC for LLM-annotator correlation [Q66, Q135]. The paper cogently justifies why general NLG metrics (BLEU/BERTScore/DialogRPT) are inadequate for pedagogical evaluation [Q4, Q7, Q8, Q10], and reports Bridge/MathDial splits separately [Q145, Q146, Q148, Q149], which is diagnostically useful. However, MRBench does not evaluate multi-turn dialogue, longer-term learning gains [Q117, Q118], code-switching handling, script appropriateness, or culturally adapted motivational output — all salient for the target deployment [WEB-1, WEB-19; cultural_norms_notes]. The output form is single-modality text: no evaluation of voice/audio output, which matters for lower-literacy learners [infrastructure_notes]. External validity for this deployment is therefore partial: the metric shape is defensible, but the coverage of authentic deployment output conditions is narrow.

ID	Evidence
Q4	“General domain-agnostic natural language generation (NLG) metrics are not well-suited for this context, as most of them fail to account for pedagogical values and require gold references, which are often not available, especially in online interactions.” (p.1)
Q7	“General domain-agnostic natural language generation (NLG) metrics like BLEU (Papineni et al., 2002), BERTScore (Lin, 2004), DialogRPT (Gao et al., 2020), and so on have been used as proxies to measure the coherence and human-likeness of AI tutor responses.” (p.2)
Q8	“However, these metrics do not account for pedagogical values (Jurenka et al., 2024; Liu et al., 2024) and often require a ground truth answer to evaluate matching responses.” (p.2)
Q10	“Additionally, these metrics can be easily manipulated; for instance, simple responses like “Hello” or “teacher:” (Baladón et al., 2023; Jurenka et al., 2024) can inflate scores.” (p.2)
Q63	“We utilize two key metrics to quantitatively assess the pedagogical effectiveness of LLMs and for comparative analysis: (1) Desired Annotation Match Rate (DAMR): This metric quantifies the percentage of responses from each human or LLM-based tutor that received the desired annotation labels.” (p.6)
Q64	“The desired labels for each dimension are detailed in Table 2.” (p.6)
Q65	“This metric offers a comparative analysis of response quality across human tutors and various LLMs, providing insights into their pedagogical performance.” (p.6)
Q66	“(2) Annotation Correlation (AC): This metric is based on Pearson’s correlation (Sedgwick, 2012), and it estimates the correlation between LLM-generated and human annotations (Kim et al., 2024), allowing us to assess the reliability of LLMs as evaluators in the context of student mistake remediation.” (p.6)
Q117	“However, one of the limitations is that it does not consider the tutoring dialogues’ impact on the overall student learning.” (p.9)
Q118	“The annotation pertains to the individual tutor turns within educational dialogues, which restricts our understanding of broader implications on student learning processes and learning gains, typically observed after a conversation concludes.” (p.9)
Q135	“The correlation scores are calculated using Pearson’s correlation (Sedgwick, 2012).” (p.13)
Q145	“Table 8: Pedagogical ability assessment of different LLMs using the DAMR scores (in %) across eight evaluation dimensions with human evaluation on the Bridge data.” (p.16)
Q146	“Table 9: Pedagogical ability assessment of different LLMs using the DAMR scores (in %) across eight evaluation dimensions with human evaluation on the MathDial data.” (p.16)
Q148	“Table 8: Pedagogical ability assessment of different LLMs using the DAMR scores (in %) across eight evaluation dimensions with human evaluation on the Bridge data. The best DAMR scores for each dimension are bolded.” (p.16)

Q149	"Table 9: Pedagogical ability assessment of different LLMs using the DAMR scores (in %) across eight evaluation dimensions with human evaluation on the MathDial data. '-' indicates that DAMR scores for Novice are not available for MathDial data. The best DAMR scores for each dimension are bolded." (p.16)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench's output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)
WEB-19	HiACC corpus documents Hindi–English code-switching in children's communication contexts (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12329218/)

Strengths

- The paper explicitly identifies limits of surface NLG metrics for pedagogical evaluation [Q4, Q7, Q8, Q10] — good conceptual grounding.
- DAMR is dimension-decomposable and diagnostically informative [Q63, Q65, Q67], and splits by dataset are reported [Q145, Q146, Q148, Q149].
- The authors explicitly flag the multi-turn / learning-gains limitation [Q117, Q118] and preliminary status of LLM-critic experiments [Q107].

ID	Evidence
Q4	"General domain-agnostic natural language generation (NLG) metrics are not well-suited for this context, as most of them fail to account for pedagogical values and require gold references, which are often not available, especially in online interactions." (p.1)
Q7	"General domain-agnostic natural language generation (NLG) metrics like BLEU (Papineni et al., 2002), BERTScore (Lin, 2004), DialogRPT (Gao et al., 2020), and so on have been used as proxies to measure the coherence and human-likeness of AI tutor responses." (p.2)
Q8	"However, these metrics do not account for pedagogical values (Jurenka et al., 2024; Liu et al., 2024) and often require a ground truth answer to evaluate matching responses." (p.2)
Q10	"Additionally, these metrics can be easily manipulated; for instance, simple responses like "Hello" or "teacher:" (Baladón et al., 2023; Jurenka et al., 2024) can inflate scores." (p.2)
Q63	"We utilize two key metrics to quantitatively assess the pedagogical effectiveness of LLMs and for comparative analysis: (1) Desired Annotation Match Rate (DAMR): This metric quantifies the percentage of responses from each human or LLM-based tutor that received the desired annotation labels." (p.6)
Q65	"This metric offers a comparative analysis of response quality across human tutors and various LLMs, providing insights into their pedagogical performance." (p.6)
Q67	"Table 3 shows DAMR scores for each LLM across all eight dimensions." (p.7)
Q107	"At the same time, we acknowledge that the experiments presented in this work are preliminary" (p.8)
Q117	"However, one of the limitations is that it does not consider the tutoring dialogues' impact on the overall student learning." (p.9)
Q118	"The annotation pertains to the individual tutor turns within educational dialogues, which restricts our understanding of broader implications on student learning processes and learning gains, typically observed after a conversation concludes." (p.9)
Q145	"Table 8: Pedagogical ability assessment of different LLMs using the DAMR scores (in %) across eight evaluation dimensions with human evaluation on the Bridge data." (p.16)
Q146	"Table 9: Pedagogical ability assessment of different LLMs using the DAMR scores (in %) across eight evaluation dimensions with human evaluation on the MathDial data." (p.16)
Q148	"Table 8: Pedagogical ability assessment of different LLMs using the DAMR scores (in %) across eight evaluation dimensions with human evaluation on the Bridge data. The best DAMR scores for each dimension are bolded." (p.16)
Q149	"Table 9: Pedagogical ability assessment of different LLMs using the DAMR scores (in %) across eight evaluation dimensions with human evaluation on the MathDial data. '-' indicates that DAMR scores for Novice are not available for MathDial data. The best DAMR scores for each dimension are bolded." (p.16)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?

OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Modality partially matches: text output is aligned with chatbot-style AI tutors on smartphones [infrastructure_notes]. However, the benchmark does not evaluate outputs in Indian English register, Hinglish, or code-switched responses [Q51; WEB-1], which are relevant to authentic deployment interactions.

OF-2: Text-to-speech / audio output is not evaluated by MRBench [Q51]. For lower-literacy or lower-English-proficiency segments of the target population [learner_profile.english_proficiency_distribution], voice-output capability may be important, and MRBench cannot assess it.

OF-3: English literacy is highly variable in the target population, with rural government-school students potentially having limited English reading/writing ability [learner_profile.english_proficiency_distribution]. MRBench's English-only text output does not accommodate lower-literacy accessibility needs.

OF-4: External-validity mismatches: (a) no evaluation of multi-turn behavior or learning gains [Q117, Q118]; (b) no evaluation of language-match appropriateness (English response to Hinglish input) [Q51, WEB-1]; (c) no evaluation of voice/audio output [Q51]; (d) DAMR is response-level and single-turn only [Q118].

ID	Evidence
Q51	"For the 192 instances in MRBench (60 from Bridge and 132 from MathDial), we generated appropriate subsequent responses based on the conversation history and the last utterance, which contained confusions or mistakes, using seven state-of-the-art LLMs." (p.6)
Q117	"However, one of the limitations is that it does not consider the tutoring dialogues' impact on the overall student learning." (p.9)
Q118	"The annotation pertains to the individual tutor turns within educational dialogues, which restricts our understanding of broader implications on student learning processes and learning gains, typically observed after a conversation concludes." (p.9)
WEB-1	Indi-RomCoM (2026): code-mixed handling is a real capability difference that MRBench's output-form evaluation does not probe (https://arxiv.org/abs/2606.30790)

Information Gaps

- Whether the deployment plans to include voice/audio output is unstated.
- The relative importance of multi-turn evaluation vs. single-turn DAMR for the deployment's operational needs is unstated.

Requires Expert Verification

- Whether Indian deployment stakeholders would consider single-turn DAMR sufficient for go/no-go decisions.

Remediation

Gap 1: MRBench evaluates only single-turn English text output; no multi-turn, no voice/audio, no learning-gains assessment [Q117, Q118].

Recommendation: Add a multi-turn evaluation protocol (extending DAMR across turns) and, if voice-first deployment is planned, a companion TTS/ASR evaluation; report separate scores for lower-English-proficiency accessibility scenarios.

ID	Evidence
Q117	"However, one of the limitations is that it does not consider the tutoring dialogues' impact on the overall student learning." (p.9)
Q118	"The annotation pertains to the individual tutor turns within educational dialogues, which restricts our understanding of broader implications on student learning processes and learning gains, typically observed after a conversation concludes." (p.9)